

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I D.Mani Kumar Reddy (192325095) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

Answer

Digital Image Processing – Image Enhancement

1. Grey Level Transformation

(a) Purpose of Grey Level Transformation in Image Enhancement

Grey level transformation is a point processing technique used to improve the appearance of an image by changing the intensity (gray level) of each pixel. It is used to:

- Improve image contrast.
- Highlight important details.
- Suppress unwanted image information.
- Prepare images for further processing.

The output pixel value depends only on the corresponding input pixel value.

(b) Negative Transformation

Formula:

$$s = 255 - r$$

Where:

- r = Input pixel value
- s = Output pixel value

Given:

$$r = [50, 100, 150, 200]$$

Calculation:

Input Pixel (r)	Formula	Output Pixel (s)
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Final Answer

$$[205, 155, 105, 55]$$

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s = c \log_{10}(1+r)$, where $c=10$, compute s for:

$r=[0,10,50,100]$

Answer

2. Log Transformation

(a) How Log Transformation Enhances Low-Intensity Pixels

Log transformation expands dark pixel values while compressing bright pixel values.

It is mainly used to:

- Enhance dark regions of an image.
- Reveal hidden details in low-intensity areas.
- Compress high-intensity values.

The mathematical expression is:

$$s = c \log(1 + r)$$

where

- c = Constant
- r = Input intensity
- s = Output intensity

(b) Calculation

Given

$$c = 10$$
$$s = 10 \log(1 + r)$$

(Using natural logarithm $\ln$)

r	Calculation	s
0	$10 \ln(1)$	0.00
10	$10 \ln(11)$	23.98
50	$10 \ln(51)$	39.32
100	$10 \ln(101)$	46.15

Final Answer

[0.00, 23.98, 39.32, 46.15]

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

Answer

3. Histogram Processing

(a) Role of Histogram Equalization

Histogram equalization is a contrast enhancement technique that redistributes pixel intensity values uniformly.

Advantages:

- Improves image contrast.
- Enhances visibility of details.
- Utilizes the full range of gray levels.
- Makes dark and bright regions clearer.

(b) Histogram Equalization

Given Histogram

Intensity	Frequency
0	2
1	2
2	4
3	8

Step 1: Total Number of Pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Since it is a **3-bit image**

$$L = 2^3 = 8$$

Gray levels = **0 to 7**

Step 2: Cumulative Distribution

Intensity	Frequency	Cumulative Frequency
0	2	2
1	2	4
2	4	8
3	8	16

Step 3: Equalized Intensity

Formula:

$$s_k = \frac{CDF}{N} \times (L - 1)$$

where

$$L - 1 = 7$$

Intensity	CDF	Calculation	Equalized Value
0	2	$(2/16) \times 7 = 0.875$	1
1	4	$(4/16) \times 7 = 1.75$	2
2	8	$(8/16) \times 7 = 3.5$	4
3	16	$(16/16) \times 7 = 7$	7

Final Answer

Original Intensity	Equalized Intensity
0	1
1	2
2	4
3	7

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

Answer

4. Spatial Filtering (Smoothing)

(a) Purpose of Spatial Filtering for Smoothing

Spatial smoothing removes noise by replacing each pixel with the average of its neighboring pixels.

Applications:

- Noise reduction.
- Image blurring.
- Preprocessing before edge detection.
- Improving image quality.

(b) Apply a 3 × 3 Averaging Filter

Given Matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Average Calculation

Sum of all values

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average

$$\frac{270}{9} = 30$$

New Center Pixel

$$\boxed{30}$$

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$

Compute the output at the center.

Answer

5. Spatial Filtering (Sharpening)

(a) How Sharpening Enhances Image Details

Sharpening emphasizes edges and fine details by increasing the difference between neighboring pixel values.

It is used to:

- Improve edge visibility.
- Highlight fine structures.
- Enhance image clarity.

The Laplacian operator is commonly used for sharpening because it detects rapid intensity changes.

(b) Apply the Laplacian Mask

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image Matrix

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Calculation

Center pixel:

$$\begin{aligned} & (4 \times 20) - (10 + 10 + 10 + 10) \\ &= 80 - 40 \\ &= 40 \end{aligned}$$

Output at the Center

40

Final Answers Summary

Question	Answer
1(b) Negative Transformation	[205, 155, 105, 55]

Question	Answer
2(b) Log Transformation	[0.00, 23.98, 39.32, 46.15]
3(b) Histogram Equalization	0→1, 1→2, 2→4, 3→7
4(b) Averaging Filter	30
5(b) Laplacian Output	40